Chapter 2

Calculating Limits Using the Limit Laws

§ Section: 2.3

2.1 Limit Laws/Laws of Algebraic Operations

Often, functions are too complex to evaluate limits directly using the precise definition. In this section, we introduce a set of simple yet powerful limit laws that make calculating limits more straightforward. Each of these laws can be rigorously proven using the precise definition of a limit.

Limit Laws

Suppose that c is a constant and the limits

$$\lim_{x \to a} f(x) \quad \text{and} \quad \lim_{x \to a} g(x)$$

exist. Then

1. $\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$
2. $\lim_{x \to a} [f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$
3. $\lim_{x \to a} [cf(x)] = c \lim_{x \to a} f(x)$
4. $\lim_{x \to a} [f(x)g(x)] = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$
5. $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$ if $\lim_{x \to a} g(x) \neq 0$

By 4

6. $\lim_{x \to a} [f(x)]^n = \left[\lim_{x \to a} f(x)\right]^n$ where n is a positive integer
7. $\lim_{x \to a} c = c$
8. $\lim_{x \to a} x = a$

9. $\lim_{x\to a} x^n = a^n$ where n is a positive integer by ④ & ⑧

10. $\lim_{x\to a} \sqrt[n]{x} = \sqrt[n]{a}$

11. $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\cancel{f(a)}}$ $= \sqrt[n]{\lim_{x\to a} f(x)}$ where n is a positive integer

[If n is even, we assume that $\lim_{x\to a} f(x) > 0$]

Direct Substitution Property

If f is a polynomial or a rational function and a is in the domain of f, then

$$\lim_{x\to a} f(x) = f(a)$$

Example 1.1

Show that $\lim_{x\to a} P(x) = P(a)$, and $\lim_{x\to a} \frac{P(x)}{Q(x)} = \frac{P(a)}{Q(a)}$ for all polynomials $P(x)$, $Q(x)$ with $Q(a) \neq 0$.

Sol: $P(x) = \sum_{k=0}^{n} C_k x^k$, where $C_k \in \mathbb{R}$.

$$\lim_{x\to a} P(x) = \lim_{x\to a} \left(\sum_{k=0}^{n} C_k x^k\right) \overset{①}{=\!=} \sum_{k=0}^{n} \lim_{x\to a} (C_k x^k)$$

$$\overset{③}{=\!=} \sum_{k=0}^{n} C_k \left(\lim_{x\to a} x^k\right) \overset{⑨}{=\!=} \sum_{k=0}^{n} C_k a^k = P(a)$$

$$\lim_{x\to a} \frac{P(x)}{Q(x)} \overset{⑤}{=\!=} \frac{\lim_{x\to a} P(x)}{\lim_{x\to a} Q(x)} = \frac{P(a)}{Q(a)}$$

Example 1.2

Compute $\lim_{x\to 1} \frac{x^2+x-6}{x^2-9}$ and $\lim_{x\to -3} \frac{x^2-9}{x+3}$.

Sol: $\because 1 \in \text{Dom}\left\{\frac{x^2+x-6}{x^2-9}\right\}$

$$\therefore \lim_{x\to 1} \frac{x^2+x-6}{x^2-9} = \frac{-4}{-8} = \frac{1}{2}.$$

$$\lim_{x\to -3} \frac{x^2-9}{x+3} = \lim_{x\to -3} x-3 = -6$$

Example 1.3

Suppose that $\lim_{x\to 2}\dfrac{f(x)-3}{x-2}=1$. Show that $\lim_{x\to 2}f(x)$ exists and find $\lim_{x\to 2}f(x)$.

Sol: $\lim_{x\to 2} f(x)-3 = \lim_{x\to 2}\left(\dfrac{f(x)-3}{x-2}\right)(x-2) = \lim_{x\to 2}\left(\dfrac{f(x)-3}{x-2}\right)\cdot \lim_{x\to 2}(x-2) = 1\cdot 0 = 0$

$\lim_{x\to 2} f(x) = \lim_{x\to 3}(f(x)-3)+3 = \lim_{x\to 3}(f(x)-3) + \lim_{x\to 3} 3 = 0+3=3.$

Example 1.4

Show that if $\lim_{x\to a}\dfrac{f(x)}{g(x)}=L$ and $\lim_{x\to a}g(x)=0$, then $\lim_{x\to a}f(x)=0$.

Exercise 1.1

Show that if $\lim_{x\to a}f(x)=L\neq 0$ and $\lim_{x\to a}g(x)=0$, then $\lim_{x\to a}\dfrac{f(x)}{g(x)}$ doesn't exist.

Properties 10 and 11 provide the foundation for evaluating limits of functions that involve radical expressions. In the simplification of such functions, the method of **rationalization** plays a central role. This technique is grounded in fundamental algebraic identities, including:

$$a^2-b^2=(a-b)(a+b)\ ,$$

$$a^3-b^3=(a-b)(a^2+ab+b^2)\ ,$$

$$a^3+b^3=(a+b)(a^2-ab+b^2)\ .$$

If $P(x)$, $Q(x)$: poly and $Q(a)\neq 0$, then

$\lim_{x\to a}\sqrt[n]{\dfrac{P(x)}{Q(x)}} = \sqrt[n]{\dfrac{P(a)}{Q(a)}}$ if $\dfrac{P(a)}{Q(a)}$ is in the domain of $\sqrt[n]{x}$.

Example 1.5 a − b

Find $\lim_{t\to0}\frac{\sqrt{t^2+9}-3}{t^2}$. $= \lim_{t\to0}\frac{(\sqrt{t^2+9}-3)(\sqrt{t^2+9}+3)}{t^2(\sqrt{t^2+9}+3)} = \lim_{t\to0}\frac{(t^2+9)-3^2}{t^2(\sqrt{t^2+9}+3)}$

$$= \lim_{t\to0}\frac{1}{\sqrt{t^2+9}+3} = \frac{1}{6}.$$

Example 1.6

Find $\lim_{x\to-2}\frac{\sqrt{x^3+10}-\sqrt{-x}}{\sqrt{x+3}-1}$. $= \lim_{x\to-2}\frac{(\sqrt{x^3+10}-\sqrt{-x})(\sqrt{x^3+10}+\sqrt{-x})(\sqrt{x+3}+1)}{(\sqrt{x+3}-1)(\sqrt{x^3+10}+\sqrt{-x})(\sqrt{x+3}+1)}$

$$= \lim_{x\to-2}\frac{(x^3+10+x)(\sqrt{x+3}+1)}{(x+2)(\sqrt{x^3+10}+\sqrt{-x})} = \lim_{x\to-2}(x^2-2x+5)\frac{\sqrt{x+3}+1}{\sqrt{x^3+10}+\sqrt{-x}}$$

$$= 13\cdot\frac{\sqrt{1}+1}{\sqrt{2}+\sqrt{2}} = \frac{13}{\sqrt{2}}$$

Example 1.7

Find $\lim_{x\to-1}\frac{x^2-1}{\sqrt[3]{x}+1}$. $= \lim_{x\to-1}\frac{(x^2-1)(\sqrt[3]{x^2}-\sqrt[3]{x}+1)}{(\sqrt[3]{x}+1)(\sqrt[3]{x^2}-\sqrt[3]{x}+1)}$

a b

$$= \lim_{x\to-1}\frac{(x^2-1)(\sqrt[3]{x^2}-\sqrt[3]{x}+1)}{x+1} = \lim_{x\to-1}(x-1)(\sqrt[3]{x^2}-\sqrt[3]{x}+1)$$

$$= -2\cdot3 = -6.$$

Example 1.8

Find $\lim_{t\to0}\frac{\sqrt{t+4}-2}{\sqrt[3]{t+1}-1}$. $= \lim_{t\to0}\frac{(\sqrt{t+4}-2)(\sqrt{t+4}+2)(\sqrt[3]{(t+1)^2}+\sqrt[3]{t+1}+1)}{(\sqrt[3]{t+1}-1)(\sqrt[3]{(t+1)^2}+\sqrt[3]{t+1}+1)(\sqrt{t+4}+2)}$

$$= \lim_{t\to0}\frac{(t+\cancel{4-4})(\sqrt[3]{(t+1)^2}+\sqrt[3]{t+1}+1)}{(t+\cancel{1-1})(\sqrt{t+4}+2)} = \lim_{t\to0}\frac{\sqrt[3]{(t+1)^2}+\sqrt[3]{t+1}+1}{\sqrt{t+4}+2}$$

$$= \frac{3}{4}.$$

2.2 Limit Laws/The Squeeze Theorem

Theorem:

If $f(x) \le g(x)$ when x is near a (except possibly at a) and the limits of f and g both exist as x approaches a, then

$$\lim_{x\to a} f(x) \le \lim_{x\to a} g(x)$$

The Squeeze Theorem

If $f(x) \le g(x) \le h(x)$ when x is near a (except possibly at a) and

$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$$

then

$$\lim_{x\to a} g(x) = L$$

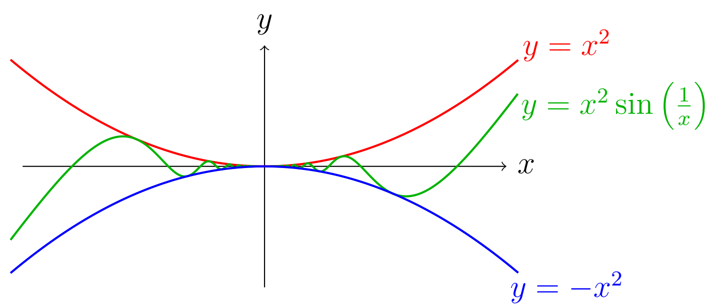
Example 2.1
Find $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right)$.

Sol: $\because$ $-1 \le \sin(\frac{1}{x}) \le 1$ for all x

$\therefore$ $-x^2 \le x^2 \cdot \sin(\frac{1}{x}) \le x^2$ for all x.

Moreover, $\lim_{x\to 0}(-x^2) = 0 = \lim_{x\to 0} x^2$.

Hence, by the Squeeze Thm, $\lim_{x\to 0} x^2 \sin(\frac{1}{x}) = 0$



Exercise 2.1

Find $\lim_{x\to 0} x\cos\left(\frac{1}{x}\right)$.

$-|x| \le x\cos(\frac{1}{x}) \le |x|$ for all x near 0.

($\boxed{-x \le x\cos(\frac{1}{x}) \le x \text{ for } x \text{ near } 0}$ is incorrect !!)

Example 2.2

Find $\lim_{x\to 0} x\left[\frac{1}{x}\right]$.

Sol: Note that $\frac{1}{x}-1 \le [\frac{1}{x}] \le \frac{1}{x}$.

For $x>0$, $x(\frac{1}{x}-1) \le x\cdot[\frac{1}{x}] \le x\cdot\frac{1}{x}$

$\Rightarrow 1-x \le x\cdot[\frac{1}{x}] \le 1$

Moreover, $\lim_{x\to 0} 1-x = \lim_{x\to 0} 1 = 1$

Thus, by the Squeeze thm, $\lim_{x\to 0^+} x[\frac{1}{x}] = 1$

For $x<0$,

$x(\frac{1}{x}-1) \ge x\cdot[\frac{1}{x}] \ge 1$

$\Rightarrow 1-x \ge x\cdot[\frac{1}{x}] \ge 1$.

Moreover, $\lim_{x\to 0^-}(1-x) = \lim_{x\to 0^-} 1 = 1$.

Thus, by the Squeeze Thm, $\lim_{x\to 0^-} x[\frac{1}{x}] = 1$

Exercise 2.2

Find $\lim_{x\to 0} \sin x\left[\frac{1}{x}\right]$.

Property:

$$\lim_{x\to a} f(x) = 0 \iff \lim_{x\to a} |f(x)| = 0$$

Example 2.3

Prove that $\lim_{x\to a} f(x) = 0$ if and only if $\lim_{x\to a} |f(x)| = 0$.

Sol: " $\Rightarrow$ "

Suppose $\lim_{x\to a} f(x) = 0$.

Then $\lim_{x\to a} |f(x)| = \lim_{x\to a} \sqrt{(f(x))^2}$

$= \sqrt{\lim_{x\to a}(f(x))^2} = \sqrt{0\cdot 0}$

$= 0$

" $\Leftarrow$ "

Suppose $\lim_{x\to a} |f(x)| = 0$.

$\because -|f(x)| \le f(x) \le |f(x)|$

and $\lim_{x\to a} -|f(x)| = \lim_{x\to a} |f(x)| = 0$.

$\therefore \lim_{x\to a} f(x) = 0$ by the Squeeze Thm.

Example 2.4

Find $\lim_{x\to 1} |x^2 - x|$.

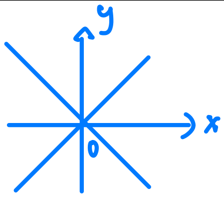
Sol: $\because \lim_{x\to 1} x^2 - x = 0$

$\therefore \lim_{x\to 1} |x^2-x| = 0$

(If $\lim_{x\to a} f(x) = 0$, then $\lim_{x\to a} |f(x)| = 0$.)

Exercise 2.3

Let $f(x) = \begin{cases} x, & \text{if } x \in \mathbb{Q} \\ -x, & \text{if } x \in \mathbb{Q}^c \end{cases}$. Find $\lim_{x\to 0} f(x)$.



$\because \lim_{x\to 0} |f(x)| = \lim_{x\to 0} |x| = 0$ $\therefore \lim_{x\to 0} f(x) = 0$.

Corollary:

If $|g(x)| \le M$ for some $M > 0$ and $\lim_{x\to a} f(x) = 0$, then $\lim_{x\to a} f(x)g(x) = 0$

Example 2.5

Prove that if $|g(x)| \le M$ for some $M > 0$ and $\lim_{x\to a} f(x) = 0$, then $\lim_{x\to a} f(x)g(x) = 0$.

Sol:

Note that $0 \le |f(x)g(x)| = |f(x)|\,|g(x)| \le M\cdot|f(x)|$.

From the assumption $\lim_{x\to a} f(x) = 0$, we have

$\lim_{x\to a} M|f(x)| = \lim_{x\to a} M\sqrt{(f(x))^2} = M\sqrt{\lim_{x\to a} f(x)} = M\cdot 0 = 0$.

Moreover, $\lim_{x\to a} 0 = 0$. Hence, by the Squeeze thm,

$\lim_{x\to a} |f(x)\cdot g(x)| = 0$.

Example 2.6

Find $\lim_{x\to 0} x\,2^{\cos\left(\frac{1}{x}\right)}$ and $\lim_{x\to 0} x\,2^{\sin\left(\frac{1}{x}\right)}$.

Sol: $\because -1 \le \cos(\frac{1}{x}) \le 1$ for all x

$\therefore \frac{1}{2} \le 2^{\cos(\frac{1}{x})} \le 2$ for all x.

Hence $\frac{|x|}{2} \le |x\,2^{\cos(\frac{1}{x})}| \le 2|x|$.

$\because \lim_{x\to 0} \frac{|x|}{2} = \lim_{x\to 0} 2|x| = 0$ $\therefore$ By the Squeeze thm,

$\lim_{x\to 0} |x\,2^{\cos(\frac{1}{x})}| = 0$.

Thus, $\lim_{x\to 0} x\,2^{\cos(\frac{1}{x})} = 0$


用直覺猜極限值
合法使用limit laws 算極限值

2.3 Using One-Sided Limits

If $f(x)$ is defined differently on two sides of the point a, then the limit of as approaches a is computed by the following theorem.

Theorem:

$$\lim_{x\to a} f(x) = L \text{ if and only if } \lim_{x\to a^-} f(x) = L \text{ and } \lim_{x\to a^+} f(x) = L$$

Example 3.1
Find $\displaystyle\lim_{x\to 0}\frac{|x^2-2x|}{3x^2+|x|}$.

Sol:

$$\lim_{x\to 0^+}\frac{|x||x-2|}{3x^2+|x|} = \lim_{x\to 0^+}\frac{x(2-x)}{3x^2+x} = \lim_{x\to 0^+}\frac{2-x}{3x+1} = 2.$$

$$\lim_{x\to 0^-}\frac{|x||x-2|}{3x^2+|x|} = \lim_{x\to 0^-}\frac{(-x)(2-x)}{3x^2-x} = \lim_{x\to 0^-}\frac{-2+x}{3x-1} = 2.$$

$$\because \lim_{x\to 0^+}\frac{|x^2-2x|}{3x^2+|x|} = \lim_{x\to 0^-}\frac{|x^2-2x|}{3x^2+|x|} = 2$$

$$\therefore \lim_{x\to 0}\frac{|x^2-2x|}{3x^2+|x|} = 2.$$